

Project Title	EV Charging Demand Prediction and Usage Pattern Analysis
Skills take away From This Project	Data Cleaning & Preprocessing Exploratory Data Analysis (EDA) Feature Engineering Regression & Classification Time Series Analysis & Forecasting Model Evaluation & Comparison Handling Class Imbalance Explainable AI – SHAP & LIME Data Visualization Python & Scikit-learn XGBoost & LightGBM Streamlit Dashboard Development
Domain	Electric Vehicles (EV) & Sustainable Transportation

Problem Statement:

Develop a machine learning system to analyze EV charging station usage patterns and predict charging demand based on factors such as charger type, location, charging company, time of charging, and historical usage behavior. The project should apply regression and classification techniques, compare multiple machine learning models, handle class imbalance, and use explainability techniques such as SHAP and LIME to understand the factors influencing charging demand. The project may also include time-series analysis to identify demand trends and forecast future charging requirements.

1. Regression – Charging Demand Prediction

Develop a regression model to **predict the charging demand of an EV charging session** using factors such as charger type, location, charger company, time of charging, and historical usage patterns. Different regression algorithms will be trained and evaluated using **MAE, MSE, RMSE, and R^2 score**.

2. Classification – High-Demand Session Prediction

Develop a classification model to **identify whether an EV charging session belongs to a high-demand or normal-demand category** based on charging and temporal characteristics. Multiple classification algorithms will be compared using **Accuracy, Precision, Recall, F1-score, and ROC-AUC**. Class-imbalance techniques such as **Random Oversampling, SMOTE, and ADASYN** can also be applied where required.

3. Time Series – Future Charging Demand Forecasting

Analyze historical EV charging demand over time to **identify trends, seasonal patterns, and variations in charging activity**, and develop time-series models to forecast future charging demand. **Moving Average and Exponential Smoothing** techniques can be applied, with forecasting performance evaluated using **MAE and RMSE**.

Business Use Cases:

1. Charging Station Demand Forecasting

Predict charging demand based on time, location, charger type, and historical usage to help operators anticipate busy and low-demand periods.

2. EV Charging Station Planning

Identify locations and time periods with consistently high demand to support decisions on where additional charging infrastructure may be required.

3. Resource and Capacity Planning

Help charging station operators plan charger availability and capacity based on expected demand and usage patterns.

4. Peak Demand Management

Identify high-demand periods and support strategies for managing charging loads and reducing congestion at charging stations.

5. Charger Utilization Analysis

Compare charging activity across different chargers, locations, and charger companies to identify utilization patterns and underused infrastructure.

6. Operational Optimization

Use predicted demand to improve staffing, maintenance scheduling, charger availability, and day-to-day station operations.

7. Customer Experience Improvement

Understanding demand patterns can help operators reduce waiting times and improve charger availability during peak periods.

8. Future Demand Forecasting

Time-series forecasting can help businesses estimate future charging demand and support long-term infrastructure and capacity planning.

Approach:

Before working on the three main problems, learners should first understand and prepare the dataset. The overall approach will be:

1. Data Understanding & Cleaning

- Understand the EV charging dataset and its features.
- Handle missing values, duplicates, and incorrect data types.
- Convert date and time columns into useful formats.

2. Exploratory Data Analysis (EDA)

- Analyze charging demand and usage patterns.
- Study demand based on charger type, location, company, and time.
- Identify trends, outliers, and important relationships.

3. Feature Engineering

- Extract features such as hour, day, month, day of week, weekend/weekday, and time of day.
- Create meaningful historical/usage-based features where appropriate.
- Prepare the final dataset for machine learning.

After this preparation, learners will work on **three separate machine learning problems**.

1. Regression – Charging Demand Prediction

Develop a regression model to predict the **charging demand** of an EV charging session.

Learners will:

- Select relevant charging, location, and time-based features.
- Build and compare different regression models.

- Apply models such as Linear Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost, and LightGBM.
- Evaluate the models using **MAE, MSE, RMSE, and R²**.
- Identify the important factors influencing charging demand.

2. Classification – High-Demand Session Prediction

Convert the demand information into categories such as **High Demand** and **Normal Demand**, based on an appropriate threshold derived from the dataset.

Learners will:

- Create the classification target.
 - **High Demand:** Demand $\geq$ median Demand
 - **Normal Demand:** Demand $<$ median Demand
- Build classification models such as Logistic Regression, Decision Tree, Random Forest, KNN, SVM, XGBoost, and LightGBM.
- Evaluate the models using **Accuracy, Precision, Recall, F1-score, and ROC-AUC**.
- Check for class imbalance.
- Apply techniques such as **Random Oversampling, SMOTE, or ADASYN** when required.
- Identify the factors associated with high-demand charging sessions.

3. Time Series – Future Charging Demand Analysis

Analyze charging demand **over time** to understand how demand changes across different periods.

Learners will:

- Convert charging timestamps into a time-series format.
- Aggregate demand by suitable time intervals such as hourly or daily.
- Analyze **trend, seasonality, and noise**.
- Check whether the time series is stationary.

- Apply **Moving Average and Exponential Smoothing**.
- Evaluate forecasting performance using **MAE and RMSE**.
- Identify periods of increasing or decreasing charging demand.

4. Streamlit Dashboard – Final Application

Finally, learners will combine the analysis and machine learning results into an interactive **Streamlit dashboard**.

The dashboard can include:

- **Overall EV charging usage KPIs**
- Charging demand by location and charger type
- Time-based demand trends
- **Regression:** predicted charging demand
- **Classification:** high-demand session prediction
- **Time Series:** demand trends and forecasting
- Model performance comparison
- Feature importance
- SHAP/LIME-based explanations
- Interactive filters for location, charger type, company, date, etc.

Final Flow:

Data Understanding → Data Cleaning → EDA → Feature Engineering → Regression → Classification → Time Series Analysis → Model Explainability → Streamlit Dashboard

Results:

By the end of the project, learners should be able to:

1. **Understand and Prepare the Dataset**
 - Clean and preprocess the EV charging dataset.

- Handle missing values, duplicates, outliers, and data types.
- Perform meaningful feature engineering from date and time information.

2. Analyze EV Charging Patterns

- Identify charging demand patterns based on location, charger type, company, and time.
- Discover important trends and relationships through EDA.

3. Build a Regression Model

- Predict EV charging demand using suitable machine learning algorithms.
- Compare model performance using **MAE, MSE, RMSE, and R²**.
- Identify important factors affecting charging demand.

4. Build a Classification Model

- Classify charging sessions into categories such as **High Demand** and **Normal Demand**.
- Handle class imbalance using suitable techniques when required.
- Evaluate models using **Accuracy, Precision, Recall, F1-score, and ROC-AUC**.

5. Perform Time Series Analysis

- Analyze charging demand over time.
- Identify **trends, seasonality, and variations** in demand.
- Apply Moving Average and Exponential Smoothing for demand forecasting.

6. Interpret Machine Learning Models

- Use **Feature Importance, SHAP, and LIME** to understand model predictions.
- Explain which features have the greatest influence on charging demand.

7. Develop an Interactive Dashboard

- Build a **Streamlit dashboard** combining EDA, predictions, forecasting, and model insights.

- Provide interactive visualizations and filters for exploring EV charging patterns.
8. **Deliver a Complete End-to-End ML Solution**
- Demonstrate the complete workflow from **raw data** → **analysis** → **machine learning** → **evaluation** → **explainability** → **deployment through Streamlit**.

Project Evaluation metrics:

- **Data Quality:** Proper data cleaning, preprocessing, and feature engineering.
- **Regression:** Evaluate demand prediction using **MAE, MSE, RMSE, and R²**.
- **Classification:** Evaluate High/Normal Demand prediction using **Accuracy, Precision, Recall, F1-score, and ROC-AUC**.
- **Time Series:** Evaluate demand trends and forecasting using **MAE and RMSE**.
- **Model Explainability:** Use **Feature Importance, SHAP, and LIME** to interpret predictions.
- **Dashboard:** Develop a functional and interactive **Streamlit dashboard** presenting insights, predictions, and forecasts.
- **Overall:** Demonstrate a complete **end-to-end ML workflow** from data preparation to deployment.

Technical Tags:

Machine Learning, Regression, Classification, Time Series Analysis, EV Charging, Demand Prediction, Feature Engineering, EDA, Scikit-learn, XGBoost, LightGBM, Random Forest, KNN, SVM, SMOTE, ADASYN, SHAP, LIME, Model Evaluation, Data Visualization, Python, Streamlit, Forecasting, Explainable AI (XAI)

Data Set:

Dataset: [Charging Record](#)

Source: Publicly available EV charging transaction dataset

Format: CSV

Records: Approximately 72,000+ charging sessions

Key Variables:

UserID, ChargerID, ChargerCompany, Location, ChargerType, StartDay, StartTime, EndDay, EndTime, StartDatetime, EndDatetime, Duration, Demand

Target Variable: Demand for regression and derived High/Normal Demand for classification.

Time Variable: StartDatetime / EndDatetime for time-series analysis.

Usage: The dataset will be used for **EDA, demand prediction, high-demand classification, time-series analysis, and Streamlit dashboard development.**

Project Deliverables:

Source Code: Complete Python/ML code with proper structure and comments.

Dataset: Processed dataset or dataset source/link.

Project Documentation: Data preprocessing, EDA, feature engineering, ML models, evaluation, and key findings.

Model Results: Regression, classification, and time-series results with evaluation metrics.

Streamlit Dashboard: Working interactive application.

Project Presentation: Brief presentation covering the approach, results, insights, and application.

README File: Project overview, setup instructions, requirements, and steps to run the application.

Project Guidelines:

- Follow **PEP 8** coding standards and use meaningful variable/function names.
- Keep the code **modular, clean, and well-commented**.
- Maintain a clear project structure for **data, notebooks, source code, models, and dashboard**.
- Use **Git/GitHub** for version control and commit changes regularly with meaningful messages.
- Maintain a **README.md** with project overview, setup steps, dependencies, and usage instructions.
- Avoid hardcoding sensitive information such as **API keys or credentials**.
- Use a **requirements.txt** file to maintain project dependencies.
- Apply proper **train-test splitting** and avoid data leakage.
- Compare multiple models and document their **evaluation metrics and results**.
- Ensure the Streamlit dashboard is **functional, user-friendly, and reproducible**.

Timeline:

The project must be completed and submitted **within 10 days from the assigned date**

References:

Project Live Evaluation	 Project Live Evaluation
EDA Guide	 Exploratory Data Analysis (ED..
Capstone Explanation Guideline	 Capstone Explanation Guideline
GitHub Reference	 How to Use GitHub.pptx
Project Submission Guidelines	 GitHub Project Submission G...
EDA	 EDA_INDEPTH
Streamlit recording (English)	 Special session for STREAML...
Pickling	 PICKLING

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

Timing: Monday-Saturday (3:00PM to 4:00PM)

Session Link:

https://docs.google.com/document/d/1J7z_Y2whgCtFDXixysXBNXqCuhXAYD9vkiNQ-7Q7XKs/edit?usp=sharing

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing:

Monday-Saturday (11:00AM to 1:00PM)

Monday-Friday (05:00PM to 06:00PM)

Booking link : <https://bit.ly/project-live-booking>